

Experiment 5

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    // Create pipe
    if (pipe(fd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // Create first child process
    pid_t pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // Child 1: execute ls -l

        close(fd[0]);          // Close read end

        dup2(fd[1], STDOUT_FILENO); // stdout -> pipe

        close(fd[1]);
```

```
    execlp("ls", "ls", "-l", (char *)NULL);

    perror("execlp ls");
    exit(1);
}

// Create second child process
pid_t pid2 = fork();

if (pid2 == -1)
{
    perror("fork");
    exit(1);
}

if (pid2 == 0)
{
    // Child 2: execute grep ".c"

    close(fd[1]);          // Close write end

    dup2(fd[0], STDIN_FILENO); // pipe -> stdin

    close(fd[0]);

    execlp("grep", "grep", ".c", (char *)NULL);

    perror("execlp grep");
    exit(1);
}

// Parent closes both pipe ends
close(fd[0]);
close(fd[1]);

// Wait for both child processes
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
```

```
    return 0;  
}
```

```
bharath@DESKTOP-6D34C2R:~/OSSP$ nano exp5_b.c  
bharath@DESKTOP-6D34C2R:~/OSSP$ gcc exp5_b.c -o exp5_b  
bharath@DESKTOP-6D34C2R:~/OSSP$ ./exp5_b  
-rw-r--r-- 1 bharath bharath 1045 Aug 25 02:54 IC.c  
-rw-r--r-- 1 bharath bharath 831 Aug 20 11:42 exp1.c  
-rw-r--r-- 1 bharath bharath 1026 Aug 20 11:44 exp2.c  
-rw-r--r-- 1 bharath bharath 873 Aug 20 11:47 exp3.c  
-rw-r--r-- 1 bharath bharath 783 Aug 20 11:49 exp4_wait.c  
-rw-r--r-- 1 bharath bharath 866 Aug 20 11:49 exp4_waitpid.c  
-rw-r--r-- 1 bharath bharath 1477 Aug 25 02:58 exp5.c  
-rw-r--r-- 1 bharath bharath 1272 Aug 25 03:04 exp5_b.c  
-rw-r--r-- 1 bharath bharath 19 Aug 20 11:54 source.txt  
-rw-r--r-- 1 bharath bharath 559 Aug 20 11:50 zombie.c  
-rw-r--r-- 1 bharath bharath 692 Aug 20 11:51 zombie_fixed.c  
-rw-r--r-- 1 bharath bharath 692 Aug 20 11:51 zombie_fixed.c  
bharath@DESKTOP-6D34C2R:~/OSSP$ ls -l | grep ".c"  
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-rw-r--r-- 1 bharath bharath 831 Aug 20 11:42 exp1.c  
-rw-r--r-- 1 bharath bharath 1026 Aug 20 11:44 exp2.c  
-rw-r--r-- 1 bharath bharath 873 Aug 20 11:47 exp3.c  
-rw-r--r-- 1 bharath bharath 783 Aug 20 11:49 exp4_wait.c  
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```